Feature selection with filter methods for an IDS flow dataset

Project "Arquitectura IDS impulsada por IA". Author: Laura Camila Herrera Murillo. All numbers in this report are read from the result files under `results/`; the scripts that produce them are listed in section 9.

1. Summary

- Three filter methods were applied to two flow datasets (A, captured with a firewall; B, without): Low Variance, Pearson correlation and Chi-square. Each filter was fitted on training rows only and each resulting feature subset was evaluated with the same RandomForest.
- No filter improves the model. What the filters achieve is a smaller input at no measurable cost: the final lists have **20 features for A** and **30 for B** (of 77) and score F1-macro 0.9678 and 0.9701 on test, against 0.9713 and 0.9669 with all features. The differences are smaller than the fold-to-fold noise measured in cross-validation (standard error about 0.0072).
- The combined pipeline (Low Variance, then Pearson, then Chi-square) reaches the baseline with fewer features than Chi-square alone (20 against 40 in A, 30 against 40 in B), because Chi-square alone fills its top ranks with redundant features.
- **The scores must not be read as attack detection ability.** The features identify which of the two data sources a row comes from almost perfectly, and labels that name the same attack in the two sources are never confused with each other. Section 8 explains this and the other limitations.

2. Data and preparation

Each dataset joins rows from an external dataset (IDS2025) with flows captured in a virtual lab, extracted with `cicflowmeter`: 79 features and the target `newLabel` with 12 labels. A and B are processed separately with the same code.

Step	A	B
Raw rows	370,545	345,192
Exact duplicates removed (after dropping the port columns)	35,062	36,079
Clean rows	335,314	308,940
Train rows (stratified, 20% test, seed 42)	268,251	247,152
Test rows	67,063	61,788

Cleaning removed the two port columns (shortcuts), rows with NaN or Inf, exact duplicates and rows with contradictory labels, leaving 77 features. Details and justifications are in `cleaning_notes.md` and `decisions.md`.

3. What each filter does

Filters score features with statistics computed from the data. They are not models and nothing is trained. All three were fitted on training rows only.

Low Variance. A feature that takes almost the same value in every row cannot separate classes. Raw variance depends on the unit of each feature, so the rule used is the share of rows taken by the most frequent value: a feature is dropped when one value covers at least 99.9% of the training rows. This removes 9 features in both datasets: the 5 constant columns and 4 flag counters with 1 to 4 non-zero rows. A looser threshold (99%) was rejected because it removes `Fwd PSH Flags` and `SYN Flag Count`, which are non-zero in 22.5% of the `Brute Force` rows: with this class imbalance a quasi-constant

feature can be the signal of a small class. The variance after MinMax scaling was computed and discarded as a rule, because it penalizes features with extreme values rather than uninformative ones (`decisions.md`, decision 6).

Pearson correlation. Two features with an absolute correlation close to 1 carry the same information. For every group of features correlated above the threshold (0.90, 0.95 and 0.99 were tried) only one is kept: the one with the highest ANOVA F score against the label. Constant features have no defined correlation, so this filter cannot remove them.

Chi-square. Measures the association between a feature and the label in a table of counts. The features are continuous, so each one is first cut into 10 quantile bins with edges learned on train; without this step the score would depend on the unit of measure. Features are ranked by the statistic and the top k are kept (k = 10, 20, 30, 40). The scores were cross-checked against the classic contingency-table test of SciPy for every feature. A second ranking by Cramer's V (the same statistic corrected for sample size and number of bins) was evaluated as a comparison.

Combined pipeline. Low Variance at 99.9%, then Pearson on the survivors, then the top k survivors by Chi-square.

4. Evaluation protocol

- Model: `RandomForestClassifier` with 200 trees and `class_weight="balanced"`, identical for every subset and not tuned.
- Metrics: F1-macro, PR-AUC macro, MCC and per-class recall. Accuracy is not reported as a main metric because the two largest classes hold about 70% of the rows.
- Settings are compared with stratified 5-fold cross-validation inside train; the filters are refitted in every fold.
- Selection rule, fixed before seeing results: within each family, the setting with the fewest features whose mean F1-macro is within one standard error of the best one.
- The test rows are used once, for the baseline and the selected setting of each family.

5. Comparison

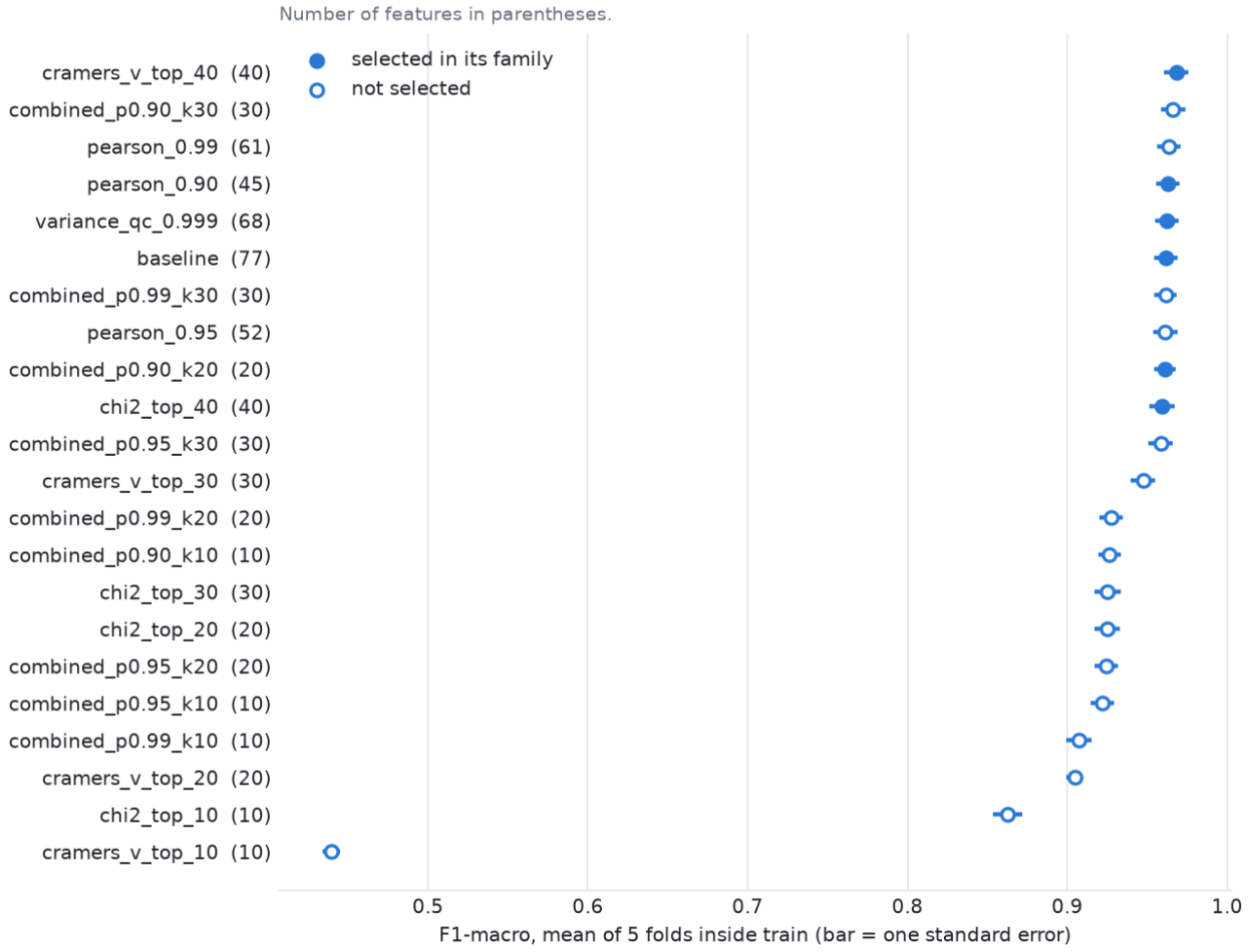
5.1 Cross-validation inside train (all configurations)

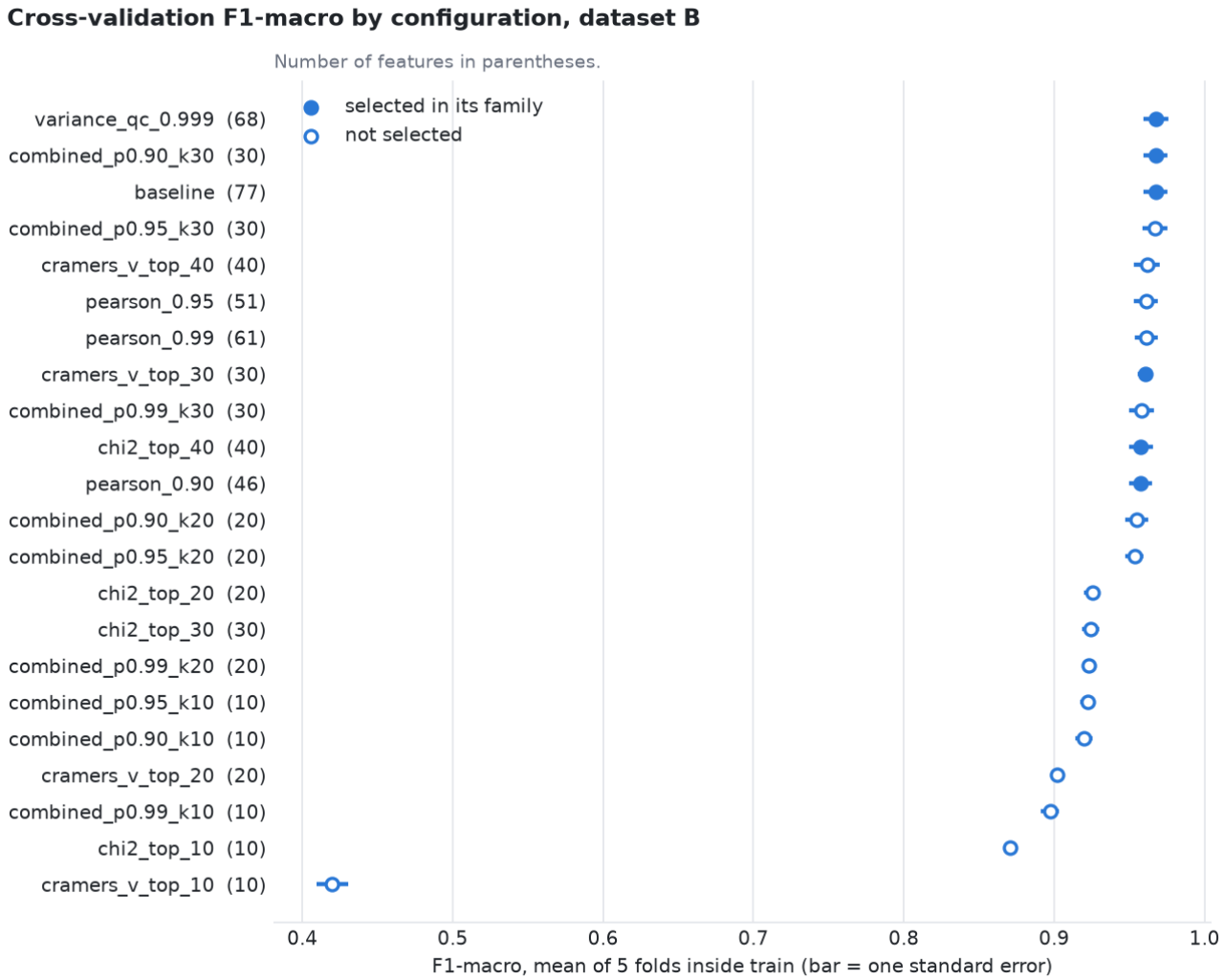
Mean of 5 folds; F1-macro is shown with its standard error. `combined_pX_kY` means Pearson threshold X and top Y by Chi-square.

Configuration	A feat.	A F1-macro	A PR-AUC	A MCC	B feat.	B F1-macro	B PR-AUC	B MCC
`baseline`	77	0.9618 ± 0.0072 **(selected)**	0.9789	0.9938	77	0.9675 ± 0.0081 **(selected)**	0.9830	0.9928
`variance_qc_0.999`	68	0.9621 ± 0.0074 **(selected)**	0.9794	0.9939	68	0.9677 ± 0.0081 **(selected)**	0.9824	0.9930
`pearson_0.90`	45	0.9630 ± 0.0075 **(selected)**	0.9792	0.9936	46	0.9574 ± 0.0076 **(selected)**	0.9753	0.9932
`pearson_0.95`	52	0.9613 ± 0.0077	0.9759	0.9942	51	0.9612 ± 0.0081	0.9827	0.9934
`pearson_0.99`	61	0.9633 ± 0.0073	0.9789	0.9940	61	0.9611 ± 0.0076	0.9805	0.9929
`chi2_top_10`	10	0.8627 ± 0.0091	0.8903	0.9618	10	0.8707 ± 0.0048	0.8823	0.9633
`cramers_v_top_10`	10	0.4396 ± 0.0056	0.5127	0.2685	10	0.4198 ± 0.0104	0.4976	0.2802
`chi2_top_20`	20	0.9250 ± 0.0079	0.9574	0.9849	20	0.9254 ± 0.0056	0.9555	0.9831
`cramers_v_top_20`	20	0.9046 ± 0.0050	0.9178	0.9731	20	0.9022 ± 0.0037	0.9139	0.9726
`chi2_top_30`	30	0.9253 ± 0.0081	0.9577	0.9862	30	0.9242 ± 0.0057	0.9555	0.9845

Configuration	A feat.	A F1-macro	A PR-AUC	A MCC	B feat.	B F1-macro	B PR-AUC	B MCC
cramers_v_top_30	30	0.9475 ± 0.0078	0.9759	0.9856	30	0.9606 ± 0.0048 **(selected)**	0.9718	0.9896
chi2_top_40	40	0.9594 ± 0.0078 **(selected)**	0.9710	0.9940	40	0.9576 ± 0.0080 **(selected)**	0.9717	0.9928
cramers_v_top_40	40	0.9682 ± 0.0075 **(selected)**	0.9800	0.9943	40	0.9618 ± 0.0085	0.9803	0.9928
combined_p0.90_k10	10	0.9266 ± 0.0067	0.9513	0.9839	10	0.9198 ± 0.0058	0.9514	0.9828
combined_p0.90_k20	20	0.9613 ± 0.0067 **(selected)**	0.9687	0.9937	20	0.9548 ± 0.0077	0.9673	0.9932
combined_p0.90_k30	30	0.9662 ± 0.0077	0.9820	0.9936	30	0.9676 ± 0.0080 **(selected)**	0.9789	0.9933
combined_p0.95_k10	10	0.9218 ± 0.0073	0.9539	0.9846	10	0.9225 ± 0.0050	0.9544	0.9814
combined_p0.95_k20	20	0.9244 ± 0.0073	0.9580	0.9859	20	0.9533 ± 0.0060	0.9702	0.9910
combined_p0.95_k30	30	0.9586 ± 0.0076	0.9705	0.9943	30	0.9672 ± 0.0083	0.9812	0.9932
combined_p0.99_k10	10	0.9072 ± 0.0078	0.9570	0.9807	10	0.8974 ± 0.0061	0.9354	0.9753
combined_p0.99_k20	20	0.9275 ± 0.0075	0.9592	0.9860	20	0.9232 ± 0.0038	0.9554	0.9832
combined_p0.99_k30	30	0.9614 ± 0.0069	0.9687	0.9942	30	0.9580 ± 0.0082	0.9734	0.9929

Cross-validation F1-macro by configuration, dataset A





5.2 Test (baseline and the selected setting of each family)

Dataset	Configuration	Features	F1-macro	PR-AUC macro	MCC	Train time (s)
A	baseline	77	0.9713	0.9688	0.9939	19.0
A	variance_qc_0.999	68	0.9721	0.9685	0.9941	21.6
A	pearson_0.90	45	0.9727	0.9685	0.9938	14.3
A	chi2_top_40	40	0.9659	0.9575	0.9940	16.0
A	cramers_v_top_40	40	0.9715	0.9691	0.9944	15.0
A	combined_p0.90_k20	20	0.9678	0.9650	0.9939	13.7
B	baseline	77	0.9669	0.9682	0.9927	19.9
B	variance_qc_0.999	68	0.9683	0.9656	0.9928	19.8
B	pearson_0.90	46	0.9692	0.9676	0.9932	15.5
B	cramers_v_top_30	30	0.9540	0.9586	0.9892	9.5
B	chi2_top_40	40	0.9621	0.9674	0.9927	19.9
B	combined_p0.90_k30	30	0.9701	0.9661	0.9932	15.7

5.3 Recall per class on test, baseline against the final list

Label	A test rows	A baseline	A final	B test rows	B baseline	B final
Botnet ARES	275	0.971	0.982	275	0.971	0.989
Brute Force	1378	1.000	1.000	1378	0.999	1.000

Label	A test rows	A baseline	A final	B test rows	B baseline	B final
Dos/DDos	4273	0.997	0.998	4273	0.998	0.998
Infiltration	6	0.667	0.667	6	0.667	0.667
PortScan	111	0.982	0.982	111	0.973	0.982
Web Attack	406	0.993	0.993	406	0.993	0.993
benign	20834	0.988	0.990	20560	0.988	0.989
bruteforce	13344	1.000	1.000	8422	1.000	1.000
dos_httpflood	25403	0.999	0.998	24706	0.999	0.999
dos_slowhttptest	165	1.000	1.000	164	1.000	1.000
recon	857	0.995	0.988	1476	0.997	0.997
sqli	11	1.000	1.000	11	1.000	1.000

Reading of the results:

- Low Variance and Pearson remove 9 to 32 features with no consistent change: the largest gap to the baseline in cross-validation is 0.010 in F1-macro (Pearson 0.90 in B, standard error 0.008) and it does not appear on test. The baseline was not being hurt by redundant features: a tree ensemble tolerates them.
- Chi-square alone needs 40 features to reach the baseline; with 10 or 20 it is clearly worse.
- The ranking by Cramer's V is not a better alternative: with 10 features its F1-macro falls to 0.44 (A), 0.42 (B) and it detects no `sqli` row.
- The largest single error of the baseline is 200 `benign` rows predicted as `Dos/DDos` in A and 203 `benign` rows predicted as `Dos/DDos` in B.

6. Why the combined pipeline beats Chi-square alone

Chi-square scores each feature on its own. Two features that are copies of each other get the same high score and both enter the top k, although the second one adds nothing. Evidence, for the same number of features:

	A (k = 20)	B (k = 30)
F1-macro in cross-validation, Chi-square top k	0.9250	0.9242
F1-macro in cross-validation, combined top k	0.9613	0.9676
Feature pairs with absolute correlation above 0.90 inside Chi-square top k	21	34
Same, inside the combined list	0	0
Features present in both the Chi-square top k and the combined list	6	13

The top of the Chi-square ranking is occupied by groups such as `Total Length of Fwd Packets` and `Subflow Fwd Bytes`, or `Fwd Packet Length Mean` and `Avg Fwd Segment Size`, which are the same quantity under two names. The combined pipeline lets Pearson keep one feature per group first, so each of the k places goes to different information.

7. Final feature lists

Chosen by the selection rule: `combined_p0.90_k20` for A and `combined_p0.90_k30` for B. The rule was not adjusted after seeing the results (`decisions.md`, decision 8), which is why the two lists have different lengths. The lists were fitted on the full training set of each dataset and are saved as `results/final_features_A.csv` and `results/final_features_B.csv`.

Dataset A (20 features)

Rank	Feature	Chi-square	Cramer's V	In the other list
1	Total Length of Fwd Packets	684,568	0.565	yes
2	Fwd Packet Length Mean	643,034	0.632	yes
3	Max Packet Length	635,350	0.628	yes
4	Fwd IAT Mean	552,886	0.479	yes
5	Fwd IAT Total	498,573	0.454	no
6	Flow Packetss	495,073	0.453	yes
7	Fwd Header Length	484,335	0.549	yes
8	Total Fwd Packets	483,129	0.548	yes
9	Flow Bytess	469,214	0.441	yes
10	Packet Length Mean	468,722	0.591	no
11	Idle Max	267,288	0.499	no
12	Idle Mean	266,483	0.498	yes
13	Idle Std	240,515	0.473	yes
14	Active Max	232,852	0.466	no
15	Active Min	206,749	0.439	yes
16	Flow IAT Min	198,537	0.430	yes
17	Init_Win_bytes_backward	198,103	0.608	yes
18	Fwd IAT Min	181,074	0.411	yes
19	Init_Win_bytes_forward	175,271	0.572	yes
20	Min Packet Length	158,262	0.768	yes

Dataset B (30 features)

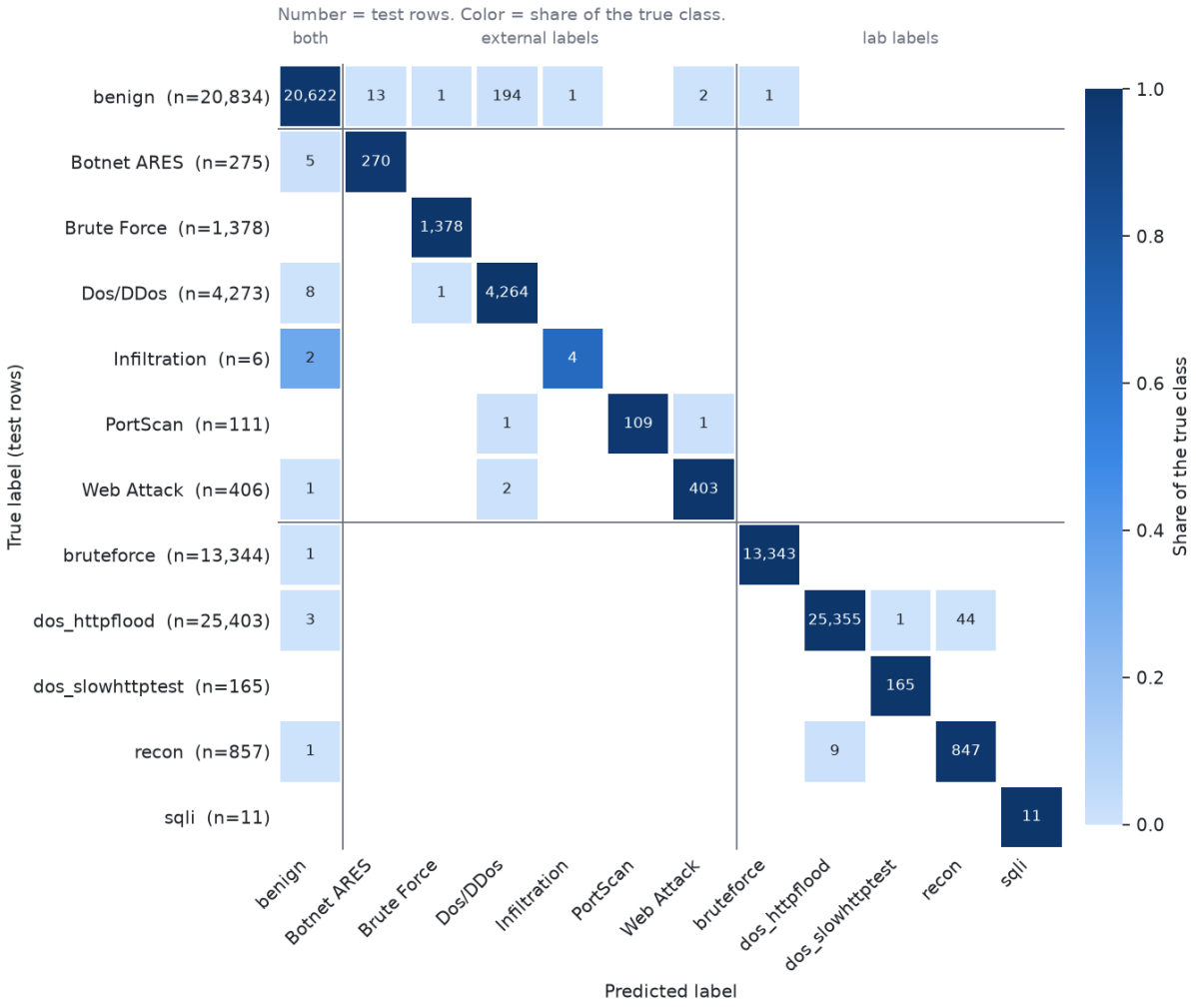
Rank	Feature	Chi-square	Cramer's V	In the other list
1	Total Length of Fwd Packets	602,592	0.552	yes
2	Average Packet Size	568,494	0.619	no
3	Fwd Packet Length Mean	494,064	0.632	yes
4	Idle Mean	488,277	0.497	yes
5	Max Packet Length	478,543	0.622	yes
6	Flow Packetss	452,778	0.451	yes
7	Flow IAT Mean	451,824	0.451	no
8	Flow Duration	445,525	0.448	no
9	Fwd IAT Mean	429,245	0.439	yes
10	Flow Bytess	411,823	0.430	yes
11	Idle Std	364,752	0.430	yes
12	Fwd Header Length	336,172	0.522	yes
13	Total Fwd Packets	288,740	0.540	yes
14	Active Min	261,817	0.515	yes
15	Active Mean	243,260	0.496	no
16	Init_Win_bytes_backward	181,329	0.606	yes
17	Flow IAT Min	174,425	0.376	yes
18	Init_Win_bytes_forward	174,182	0.594	yes
19	Min Packet Length	143,281	0.761	yes
20	Fwd Packet Length Min	143,193	0.761	no
21	Bwd IAT Total	142,165	0.758	no

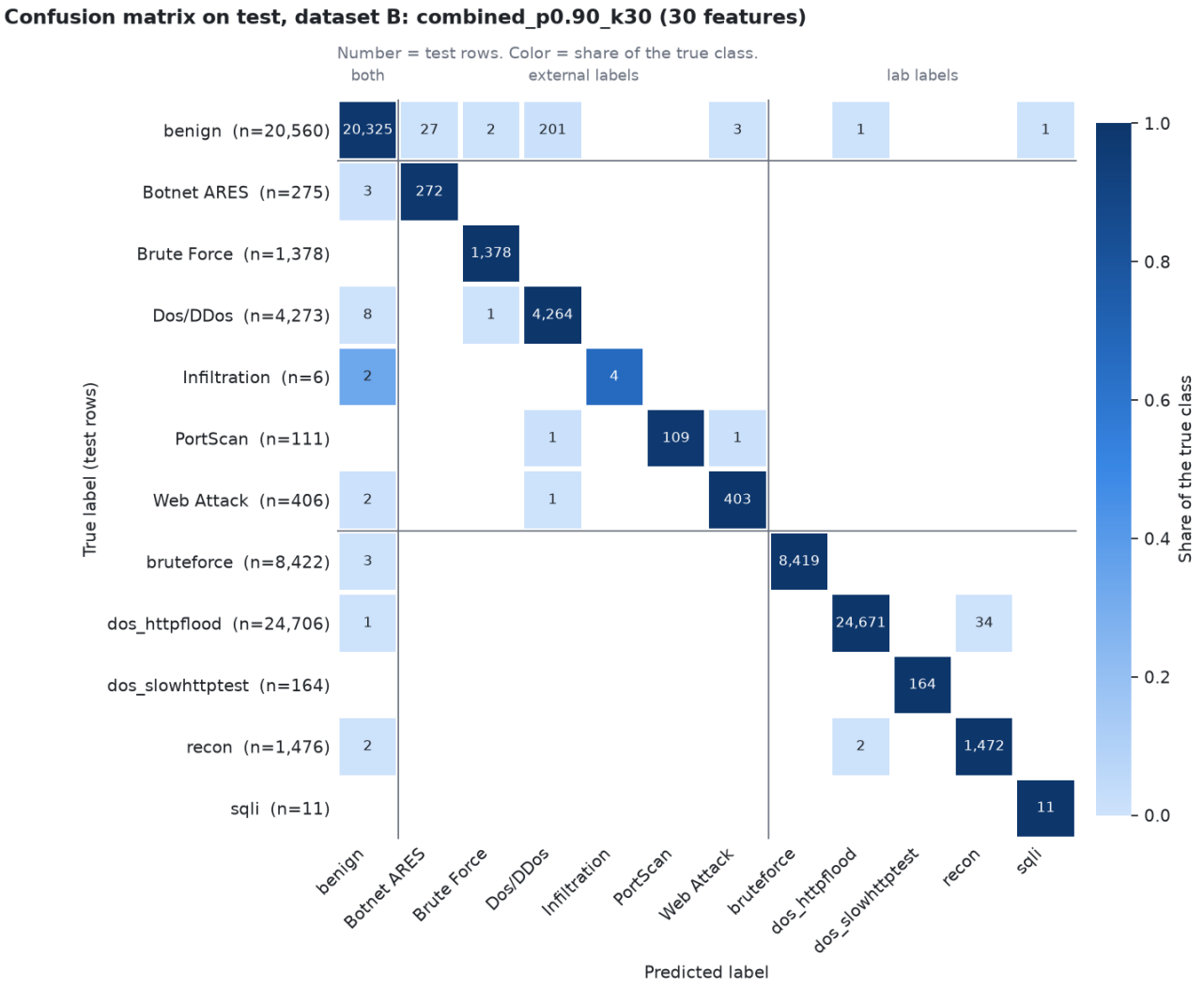
Rank	Feature	Chi-square	Cramer's V	In the other list
22	Bwd IAT Mean	141,991	0.758	no
23	Subflow Bwd Bytes	124,241	0.709	no
24	Bwd IAT Min	118,020	0.691	no
25	Bwd Packet Length Mean	106,954	0.658	no
26	Fwd IAT Min	91,347	0.304	yes
27	Bwd Packetss	83,502	0.581	no
28	Down Up Ratio	66,174	0.517	no
29	Protocol	61	0.016	no
30	Bwd Packet Length Min	0	0.000	no

A against B

- Shared (16): `Active Min`, `Flow Bytess`, `Flow IAT Min`, `Flow Packetss`, `Fwd Header Length`, `Fwd IAT Mean`, `Fwd IAT Min`, `Fwd Packet Length Mean`, `Idle Mean`, `Idle Std`, `Init_Win_bytes_backward`, `Init_Win_bytes_forward`, `Max Packet Length`, `Min Packet Length`, `Total Fwd Packets`, `Total Length of Fwd Packets`
- Only in A (4): `Active Max`, `Fwd IAT Total`, `Idle Max`, `Packet Length Mean`
- Only in B (14): `Active Mean`, `Average Packet Size`, `Bwd IAT Mean`, `Bwd IAT Min`, `Bwd IAT Total`, `Bwd Packet Length Mean`, `Bwd Packet Length Min`, `Bwd Packetss`, `Down Up Ratio`, `Flow Duration`, `Flow IAT Mean`, `Fwd Packet Length Min`, `Protocol`, `Subflow Bwd Bytes`

Confusion matrix on test, dataset A: combined_p0.90_k20 (20 features)





In the confusion matrices the labels are ordered by data source. Every misclassification falls inside one source block or involves `benign`, the only label present in both sources.

8. Limitations

8.1 The features identify the data source

External rows and lab rows come from different environments and different flow extractors, and the label vocabularies differ (`Brute Force` and `bruteforce`, `PortScan` and `recon`, `Web Attack` and `sqli`). Two checks were made:

- **Confusions between equivalent labels:** 0 in total, across the 12 configurations scored on test, in either direction. If the features described the attack itself, flows of the same attack type from the two sources would be confused at least occasionally.
- **Source probe:** a RandomForest with the same settings was trained to answer only "external or lab?" using `benign` rows alone, the one label present in both sources. With the final lists it misclassifies 0 of 20,834 test rows in A and 2 of 20,560 in B.

Part of the reason is visible in the features: 27 of the 77 features are zero in every lab row (see 8.3), so any non-zero value marks an external row. The final list of A contains 1 of them (`Init_Win_bytes_backward`) and the final list of B contains 9 (`Init_Win_bytes_backward`, `Bwd IAT Total`, `Bwd IAT Mean`, `Subflow Bwd Bytes`, `Bwd IAT Min`, `Bwd Packet Length Mean`, `Bwd Packetss`, `Down Up Ratio`, `Bwd Packet Length Min`).

Consequence: the model separates the sources and then the classes inside each source. The high scores do not show that the selected features characterize an attack type, and they say nothing about traffic from another network. The experiment that would test it, training on one source and evaluating on the equivalent labels of the other, was not done.

8.2 Infiltration

`Infiltration` has 29 rows in total: 23 in train and 6 in test. On test the model detects 4 of 6 in A and 4 of 6 in B, in every configuration; one row is 17 points of recall. In cross-validation, over the 23 training rows, recall ranges from 0.57 to 0.70 in A and from 0.52 to 0.70 in B depending on the configuration. No conclusion about which features help this class can be drawn from so few rows, and it is the class that limits F1-macro. `sqli` has 11 test rows after deduplication, so its recall is also indicative only.

8.3 One-direction flows

In 100% of the lab rows the backward packet count is 0 and the flow source is the server. The client side of every lab conversation is missing: no attacker requests and no SQL payloads are represented. Another extractor (NTLFlowLyzer) produces bidirectional flows from the same capture, so the extraction failed, not the capture. All backward features are constant for lab rows, and the lab classes are being separated only by what the server sent.

8.4 Duplicates

With the port columns present the raw files have 94 (A) and 100 (B) exact duplicate rows. Once the ports are dropped, 35,062 (A) and 36,079 (B) rows are identical to another row and were removed, keeping one copy. The effect is concentrated: `PortScan` goes from 25,409 rows to 554 and `sqli` from 179 to 58 (A) and 57 (B). The metrics therefore describe performance per distinct flow pattern, not per captured flow, and class frequencies no longer reflect the captured traffic (`decisions.md`, decision 4). Rows that are almost identical were not removed and can fall on both sides of the train/test split, which may make test scores optimistic.

8.5 Other limits

- **Chi-square and sparse features.** With quantile bins, a feature where one value covers about 90% of the rows or more collapses into a single bin and gets a score of 0 whatever information it carries. This happens to 17 non-constant features in A and 16 in B, mostly flag counters. In B only 39 features survive Pearson, so the top 30 takes most of them and includes one feature with a score of 0 (`Bwd Packet Length Min`).
- **One model.** The comparison uses a RandomForest. Models that are sensitive to redundant features may benefit more from the filters.
- **Five folds.** The standard error used by the selection rule is itself a rough estimate.
- **A and B share their external rows,** so results on the two datasets are not independent replications.

9. Reproducibility

Run in order with `.venv/bin/python`:

Script	Stage
`src/00_setup.py`	Load and check the raw files
`src/01_audit.py`	Quality audit
`src/02_clean.py`, `src/02b_duplicates_check.py`	Cleaning and duplicate check
`src/03_split.py`, `src/03b_handoff.py`	Train/test split and data package
`src/04_filters.py`	The three filters on the full training set
`src/05_evaluate.py`, `src/05b_plots.py`	Evaluation and figures

Script	Stage
`src/06_deliverables.py`, `src/06b_report.py`	Final lists and this report

Parameters are in `src/config.py`. Seeds are fixed. Stages 1 to 4 and 6 were run twice and produced identical files; stage 5 (about one hour) was run once.